



Mark Scheme (Results)

January 2025

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H/1B)
Unit 1: Principles and Applications of Science I
(Biology)

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Unit 1: Principles and Applications of Science I (Biology)

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1(a)	A depolarization		1
1(b)	sodium / Na ⁺	accept Na	1
1(c)	B –55 mV		1
1(d)	increases (the speed of an action potential through an axon) (1) (myelination) enables saltatory conduction / the impulse can jump between the nodes (of Ranvier) (1)		2
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
2(a)	conversion (1) 0.0128 mm OR 3200 µm substitution (1) $3.200 \div 0.0128$ OR $3200 \div 12.8$ evaluation (1) (x) 250	250 alone gains all 3 marks	3
2(b)	An explanation linking: (ciliated epithelial cells) require high levels of ATP (1) to move their cilia / to (re)move mucus / debris / pathogens (1) mitochondria {are the site of (aerobic) respiration / produce ATP/ release energy} (1)	accept description, such as waft mitochondria produce high levels of ATP gains 2 marks	3
Total = 6 marks			

Question Number	Answer	Additional Guidance	Mark
3(a)	protein synthesis / translate mRNA into proteins	accept polypeptide for protein	1
3(b)	Any two from: circular (1) piece of (recombinant) DNA (1) double stranded (1)	ignore small accept genetic material	2
3(c)	An explanation linking: Gram-positive bacteria decrease (1) the antibiotic only affects Gram- positive (bacterial infections) (1) antibiotics {disrupt / damage} the {cell wall / peptidoglycan} (1) Gram-positive bacteria do not have an outer membrane that protects the cell wall (1)	accept the antibiotic has no effect on the Gram- negative bacteria accept Gram-negative bacteria has an outer membrane that protects the cell wall	4
Total = 7 marks			

Question Number	Answer	Additional Guidance	Mark
4(a)	(cell wall) is made of cellulose / inelastic (1) plasma membrane can only expand to the limit of the cell wall (so the cell will not burst) (1)	accept rigid accept cell to withstand pressure	2
4(b)	Any four from: pushes chloroplasts to the edge of the (palisade) cell (1) {increases / maximises} light absorption (1) {decreases / minimises} the diffusion pathway (1) increasing {carbon dioxide (absorption) / exchange of gases} (for the chloroplasts) (1)	accept chlorophyll	4
Total = 6 marks			

Question Number	Indicative Content
5	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content, but learners should be rewarded for other relevant answers.</i> increasing serotonin levels in the synapse decreases the symptoms of depression serotonin cannot pass through the blood-brain barrier drug cannot be serotonin act on the pre-synaptic membrane a drug that increases the rate of serotonin release from the vesicles at the presynaptic membrane precursor of serotonin drug could be a precursor to serotonin drug can therefore pass through the blood-brain barrier and then be converted into serotonin therefore, could increase the levels of serotonin in the brain prevent breakdown of serotonin drug could inhibit enzymes which cause the breakdown of serotonin so more serotonin stays in the synapse inhibit reuptake drug could be a selective serotonin reuptake inhibitor (SSRI) SSRIs inhibit the reuptake pump in the presynaptic membrane more serotonin remains in the synaptic cleft more impulses travel along the post-synaptic axon agonist drug could be an agonist an agonist is a substance that mimics the actions of neurotransmitter a serotonin agonist would produce a response when it binds to a specific receptor in the brain

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Adequate interpretation, analysis and / or evaluation of the scientific information with generalised comments being made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The discussion shows some structure and coherence.
Level 2	3–4	• Good analysis, interpretation and / or evaluation of the scientific information. • Lines of argument mostly supported through the application of relevant evidence. • The discussion shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Comprehensive analysis, interpretation and / or evaluation of all pieces of scientific information. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The discussion shows a well-developed structure which is clear, coherent and logical.
		Total = 6 marks



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